

A circular wreath of various botanical illustrations surrounds a central white circle. The plants include green ferns, orange flowers, red leaves, purple flowers, and green leaves with white veins.

VP150 RC7

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Overview

Equilibrium

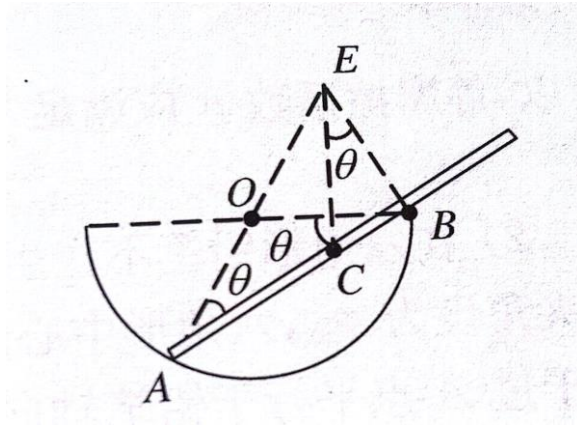
Elasticity

Fluid

Gravitation



A bowl fixed to the horizontal plane of radius R is frictionless. A frictionless stick of length L has one side in the bowl and the other side out. When equilibrium has reached, what is θ ?





$$\text{elastic modulus} = \frac{\text{stress}}{\text{strain}}$$

where stress is force per unit area, and strain is the fractional deformation due to the stress.

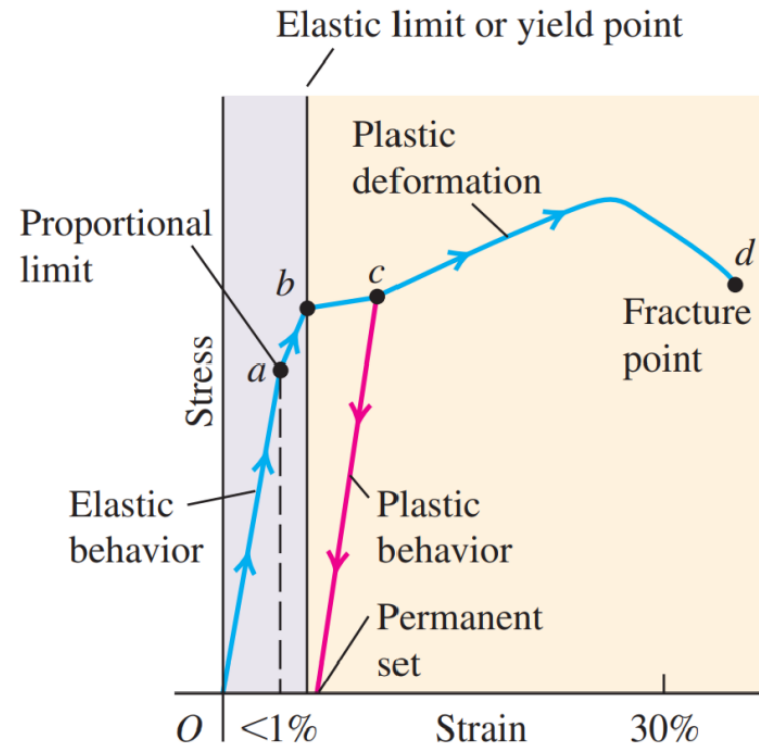
$$\text{Young's Modulus (tensile stress)} \quad Y = \frac{F_{\perp}/A}{\Delta l/l_0}$$

$$\text{Bulk Modulus (compressive stress)} \quad B = \frac{-\Delta p}{\Delta V/V_0}$$

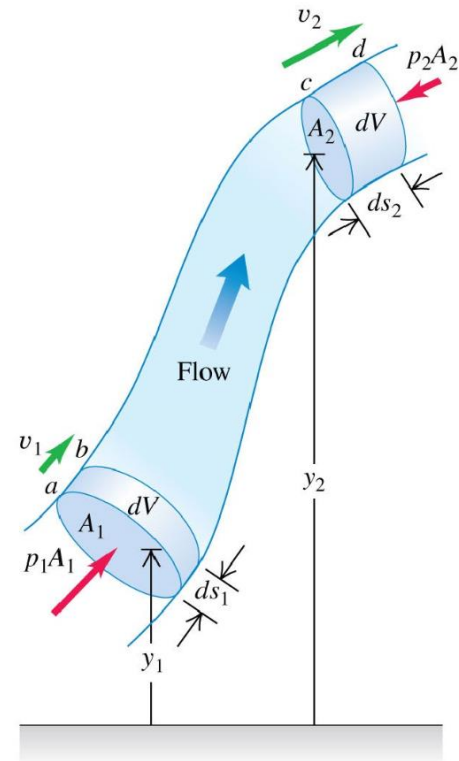
$$\text{Shear Modulus (shear stress)} \quad S = \frac{F_{\parallel}/A}{x/h} = \frac{F_{\parallel}/A}{\sin \theta}$$

*Shear Modulus is usually much smaller.



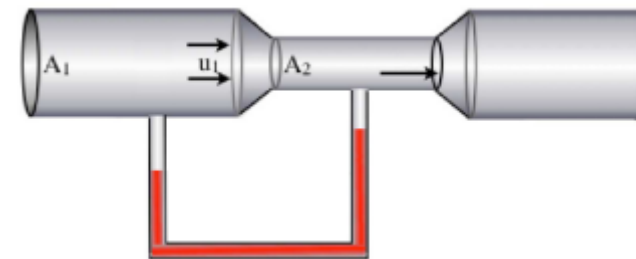


$$p + \frac{1}{2}\rho v^2 + \rho gy = \text{const}$$



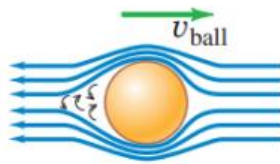


The wide section of a pipe (with cross-sectional area A_1) in the apparatus shown below transports an incompressible fluid of density ρ with an unknown flow speed. The wide and narrow section (of cross-sectional area A_2) of the pipe are connected via a U-shaped pipe that is filled with another incompressible fluid (marked in red) of density $\rho_f > \rho$. How can this apparatus be used to determine the flow speed in the wider section of the tube? Give an expression for the speed in terms of A_1 , A_2 , ρ , ρ_f , the height of the fluid in the U-pipe and the acceleration due to gravity g .





(a) Motion of air relative to a nonspinning ball



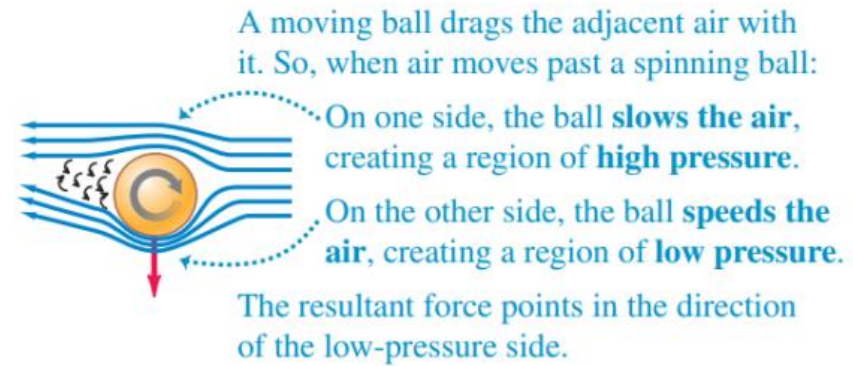
(b) Motion of a spinning ball

This side of the ball moves opposite to the airflow.



This side moves in the direction of the airflow.

(c) Force generated when a spinning ball moves through air





Meniscus and capillarity

adhesion forces dominate – concave liquid surface

cohesion forces dominate – convex liquid surface

Pressure

$$dp = -\rho gy$$

$$\Delta p = \int_{p_1}^{p_2} dp$$

*The density of mass ρ and the gravity acceleration g might be function of depth y .





(12 points) Consider a solid sphere of water sitting in deep space. The sphere has a uniform density of $\rho = 1000 \text{ kg/m}^3$ and a radius of $R = 100 \text{ km}$.

Using what we know about gravity, find a formula for the force of gravity everywhere within the volume of the sphere. Find the pressure of the water at a distance $r = 1 \text{ meter}$ from the center of the sphere.





First cosmic velocity:↵

$$G \frac{M_E m}{R_E^2} = m \frac{v_1^2}{R_E} \leftarrow$$

$$v_1 = \sqrt{\frac{GM_E}{R_E}} \leftarrow$$

Second cosmic velocity:↵

$$\frac{1}{2} m v_2^2 - G \frac{M_E m}{R_E} = 0 \leftarrow$$

$$v_2 = \sqrt{\frac{2GM_E}{R_E}} \leftarrow$$

Third cosmic velocity:↵

$$E_{k1} = \frac{1}{2} m v_2^2 \leftarrow$$

$$v_{ES} = \sqrt{\frac{GM_S}{r}} \leftarrow$$

$$v_{OS} = \sqrt{\frac{2GM_S}{r}} \leftarrow$$

$$v = v_{OS} - v_{ES} \leftarrow$$

$$\frac{1}{2} m v_3^2 = \frac{1}{2} m v_2^2 + \frac{1}{2} m v^2 \leftarrow$$

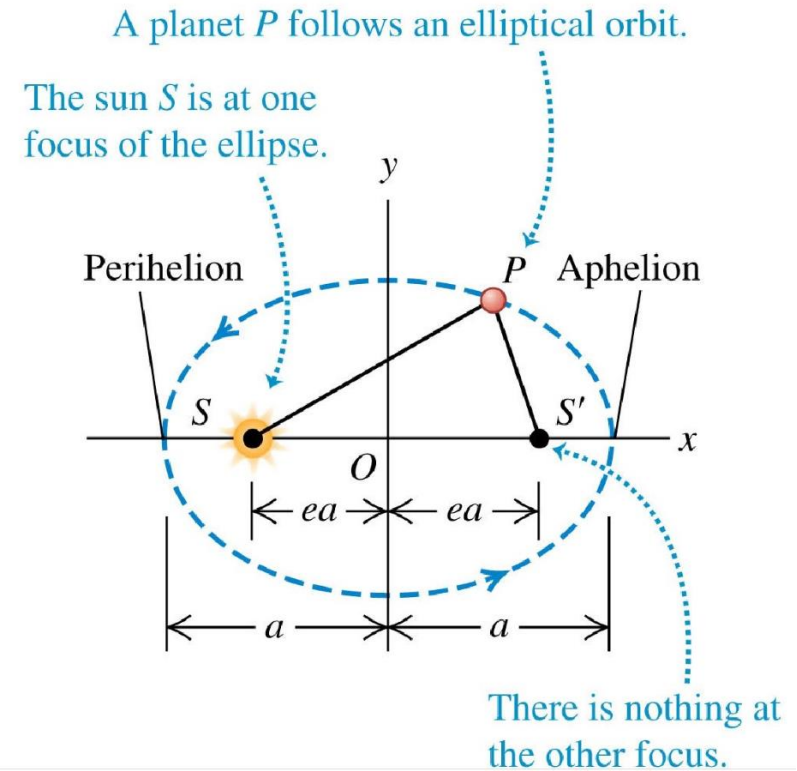
$$v_3^2 = \left(\sqrt{\frac{2GM_S}{r}} \right)^2 + ((\sqrt{2} - 1) \sqrt{\frac{GM_S}{r}})^2 \leftarrow$$





First law

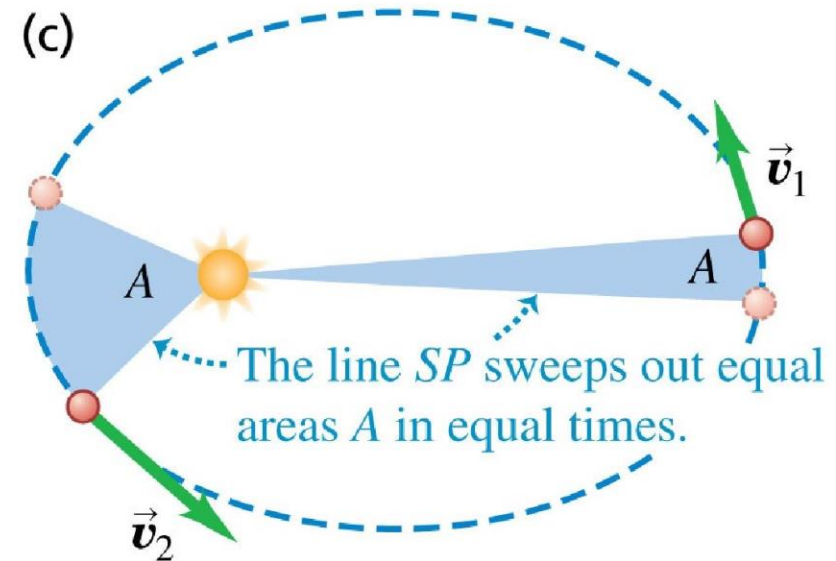
Planets move around the sun in an elliptical orbit, where the sun occupies a focus point of the ellipse





Second law

The area covered by the motion of the planets during equal time intervals must be the same.





Third law

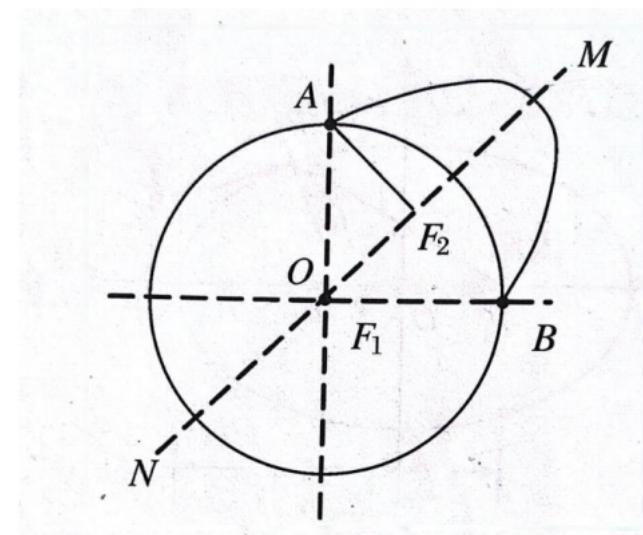
The period of motion of a planet is proportional to the 3/2-power of the major axis length of its orbit

$$T = 2\pi \frac{a^{3/2}}{\sqrt{GM}}$$





A missile is launched at the north pole of Earth and land at the equator. What is the least launching speed? Suppose the radius of Earth is R_E and the mass of Earth is M_E





Reference

Jingjia Peng

- VP150 recitation class, 2023

Zhemin Qu

- VP150 recitation class, 2023



Thanks



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冰岛



扫一扫上面的二维码图案，加我为朋友。